SEMESTER III MINI PROJECT

PlacementRadar

System diagrams: architecture, flows, data model and project structure

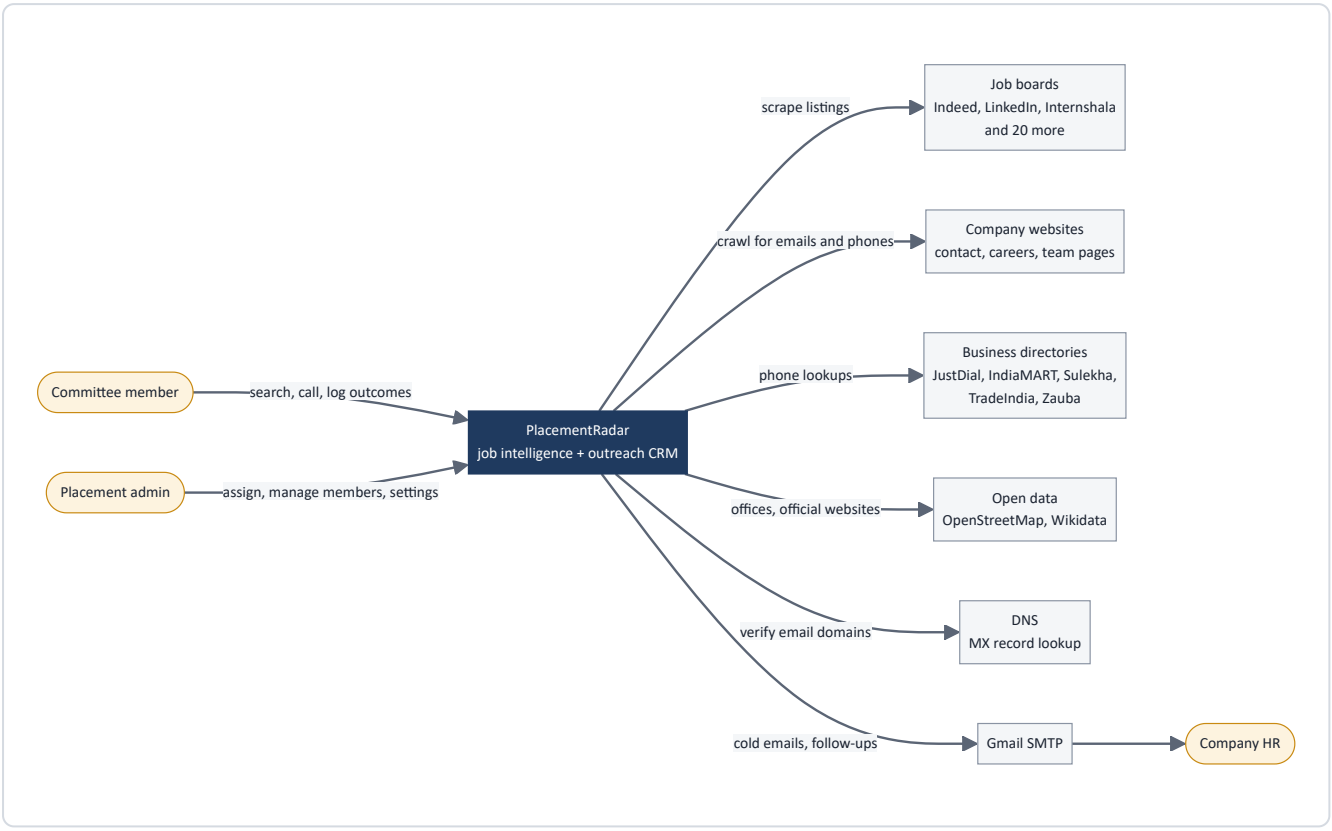
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Companion to	PROJECT_FINAL_REPORT

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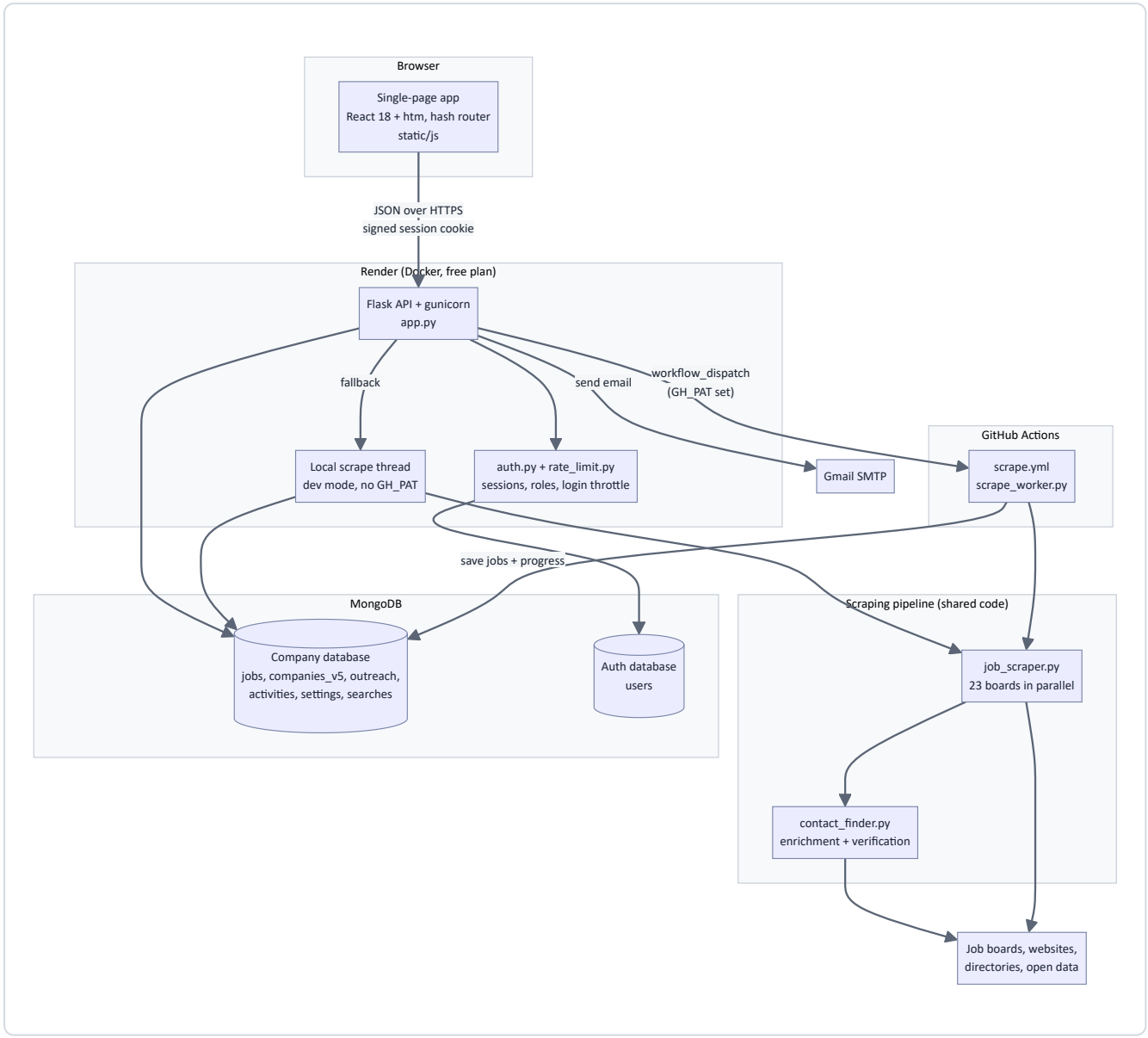
1 System context

Who uses PlacementRadar and which outside systems it talks to.



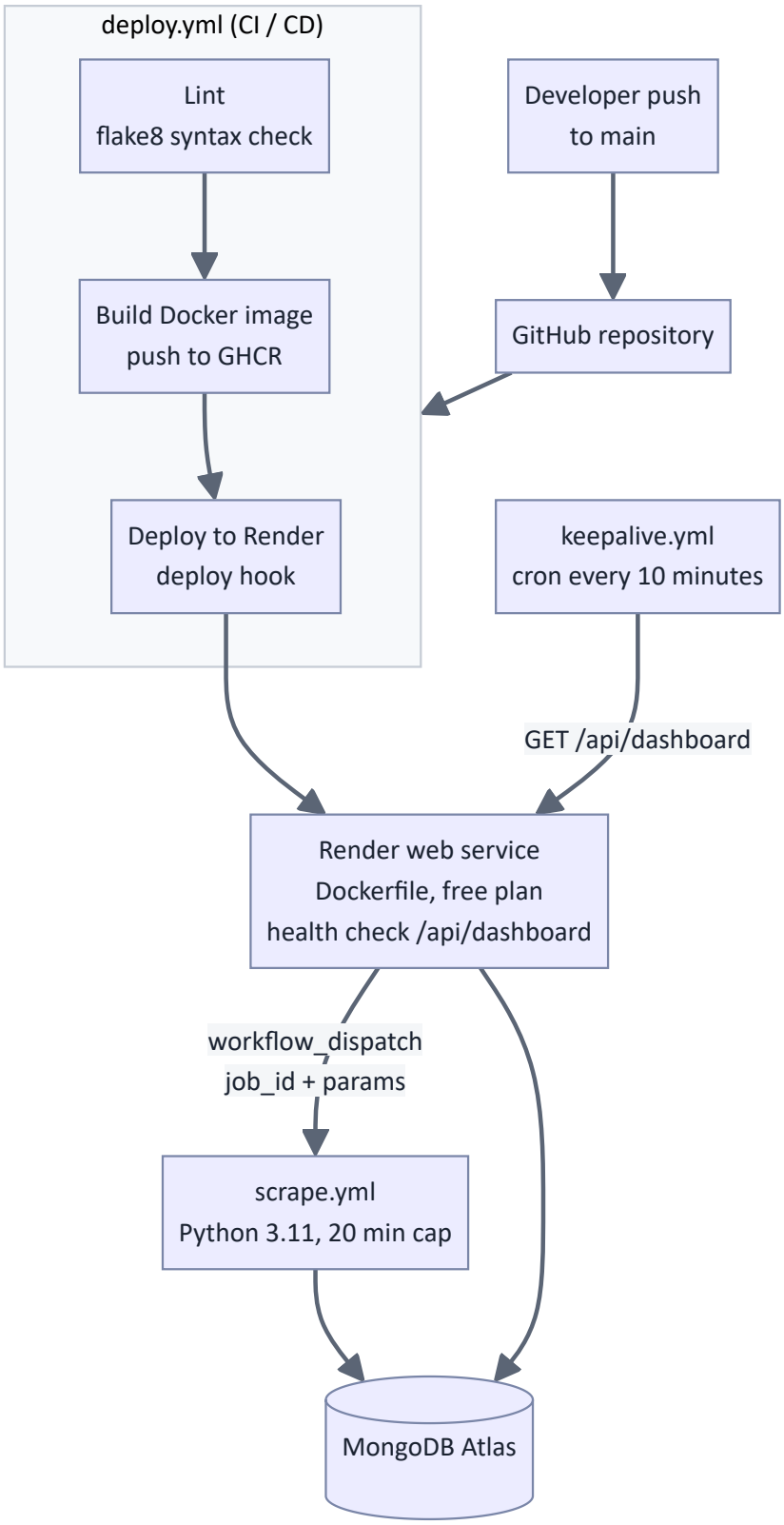
2 Container architecture

The running pieces. Heavy scraping runs on GitHub Actions in production, so the free web server stays light; without a GitHub token it falls back to a local thread.



3 Deployment and CI/CD

From a push to a running service, plus the two scheduled or on-demand workflows.



The free Render plan sleeps when idle; the keepalive workflow pings the health endpoint every 10 minutes to keep it awake.

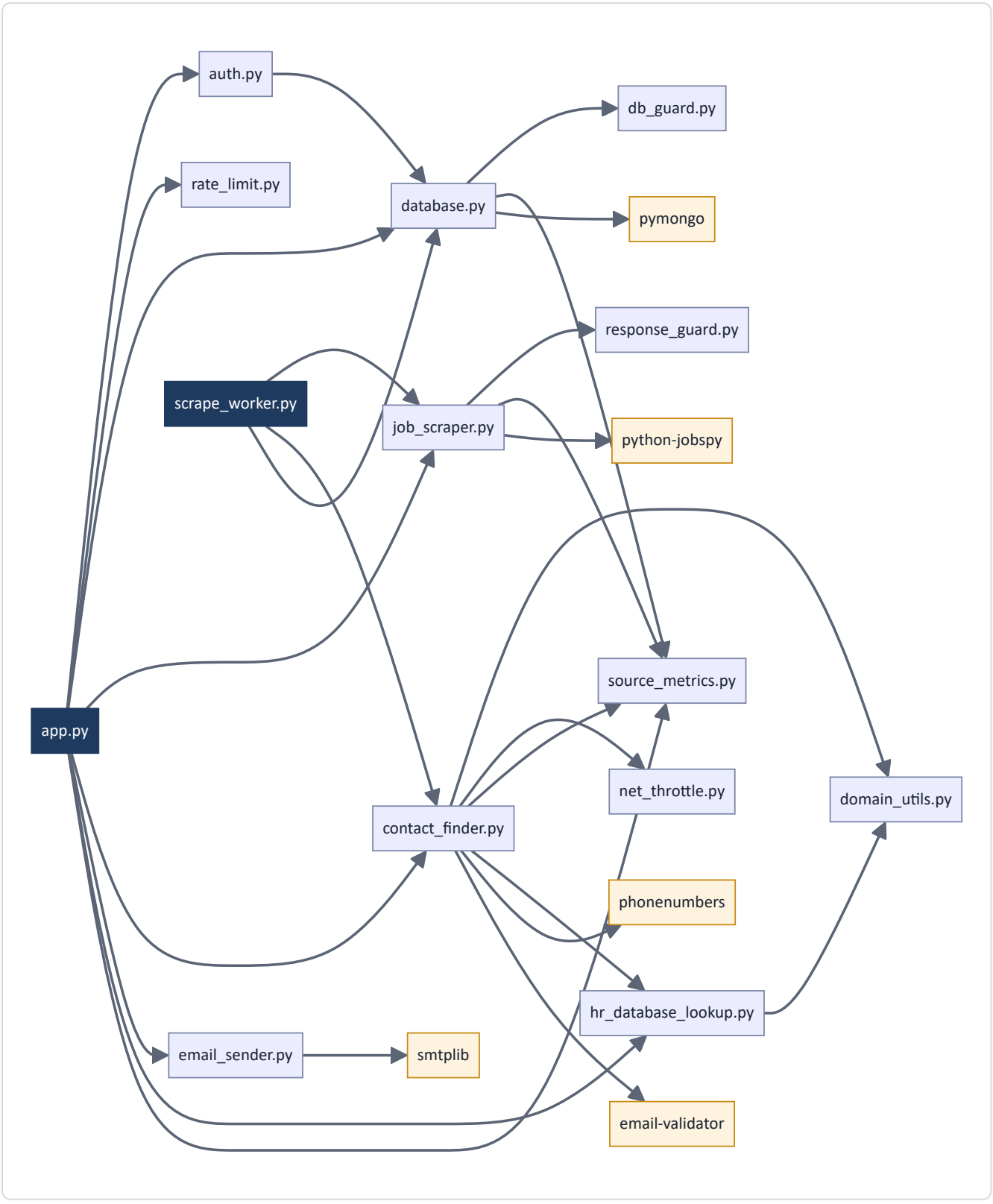
4 Project structure

Main folders and files, with what each one is for.

```
placement_tool/
├── app.py                  Flask app: 40 API routes, scrape dispatch, outreach
├── auth.py                 sign-in, sessions, member/admin roles
├── rate_limit.py           sliding-window login throttle
├── database.py             MongoDB access, pipeline, scoring, contacts
├── db_guard.py             write guards: target check + restore dump
├── job_scraper.py          23 job boards, JobSpy, parallel run, tech filter
├── contact_finder.py       enrichment: sites, directories, verification
├── domain_utils.py         company keys, domains, corroboration gate
├── response_guard.py       catches redirects and off-topic HTTP 200 pages
├── net_throttle.py         per-domain politeness: gap, backoff, UA
├── source_metrics.py       per-source health counters
├── email_sender.py         Gmail SMTP sending
├── hr_database_lookup.py   committee HR directory lookups
├── scrape_worker.py        entry point for the GitHub Actions scraper
├── templates/
│   ├── index.html          22-line shell for the single-page app
│   └── index_legacy.html   previous 2,700-line page, kept for reference
├── static/
│   ├── js/ app.js lib.js ui.js  shell, helpers, shared components
│   │   └── views/                today, board, pipeline, search, outreach,
│   │                             hr, sites, settings, admin, presets
│   ├── css/ tokens.css app.css
│   ├── fonts/                  Atkinson Hyperlegible Next
│   └── vendor/                 React, ReactDOM, htm (pinned, offline)
├── tools/                      17 one-off pipelines: bench, domain resolution,
│                                 verification, OSM offices, HR import, review sheets
├── tests/                      28 pytest tests, no network or database
├── .github/workflows/          deploy.yml, scrape.yml, keepalive.yml
├── Dockerfile  docker-compose.yml  render.yaml
├── requirements.txt
└── README.md  DOCUMENTATION.md  PROJECT_FINAL_REPORT.*
```

5 Backend module dependencies

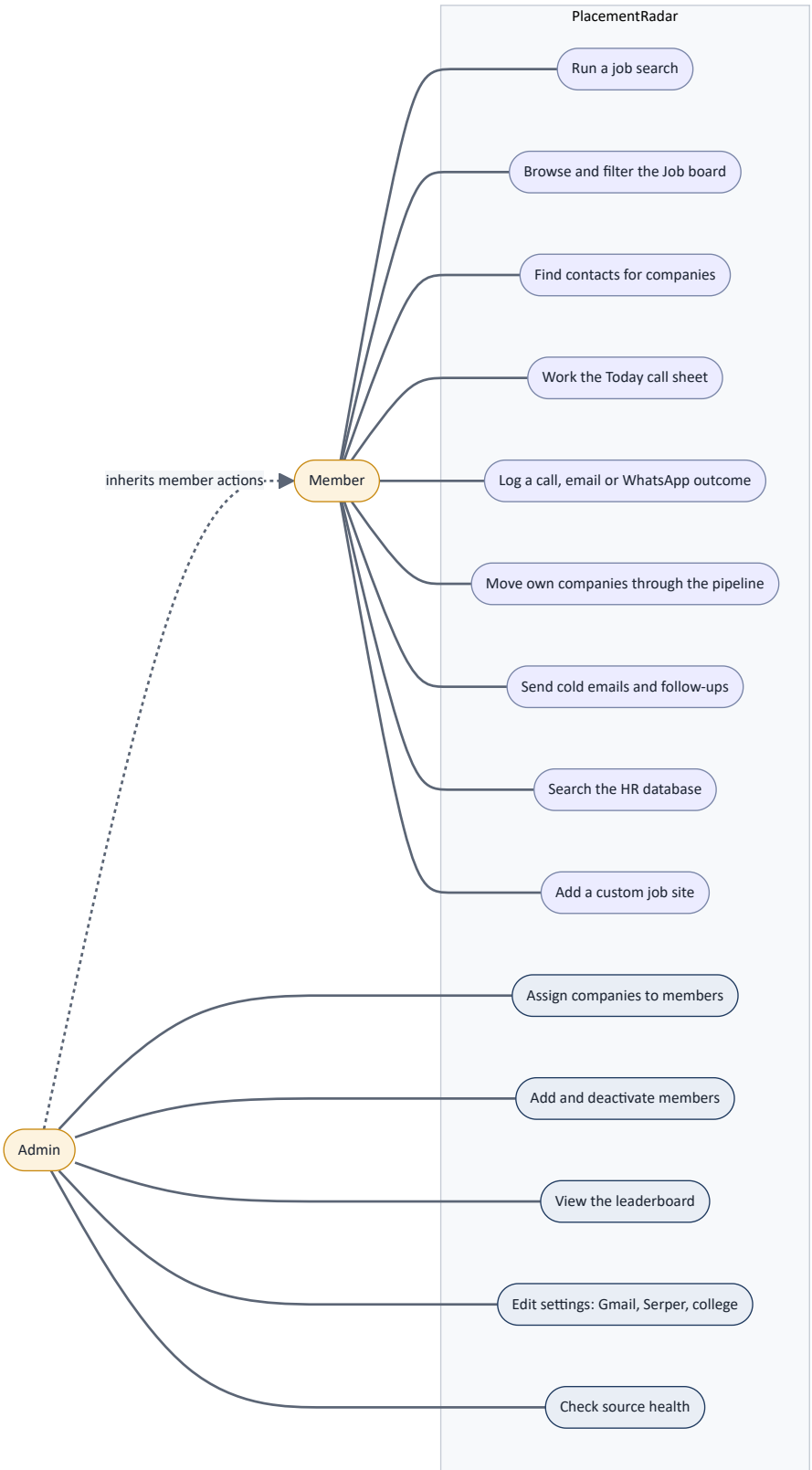
Which Python module imports which. Arrows point from the importer to the module it uses.



source_metrics, net_throttle and domain_utils import no database code, so the benchmark harness can run them without touching MongoDB.

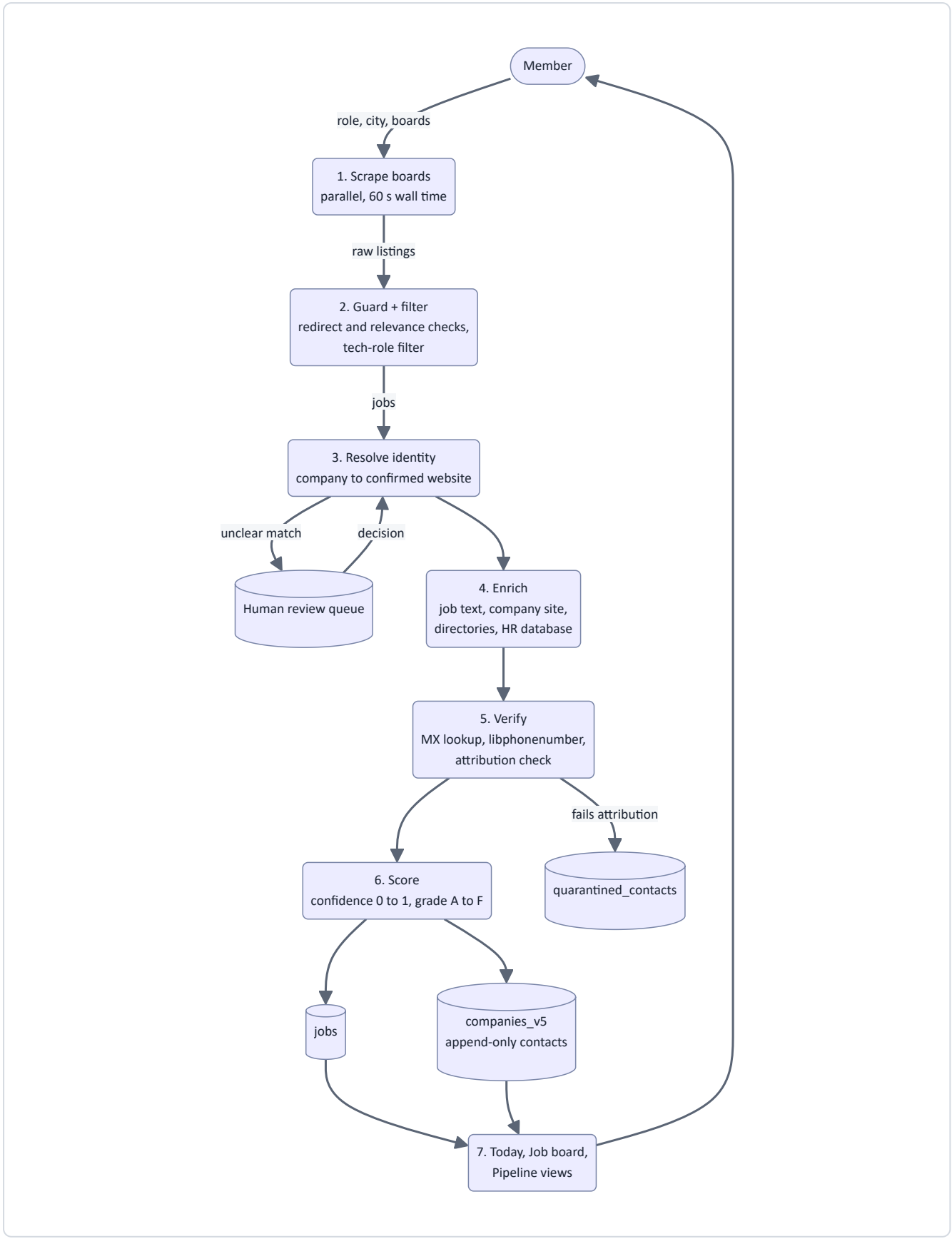
6 Use cases by role

What each role can do. Admins can do everything a member can, plus the admin-only actions.



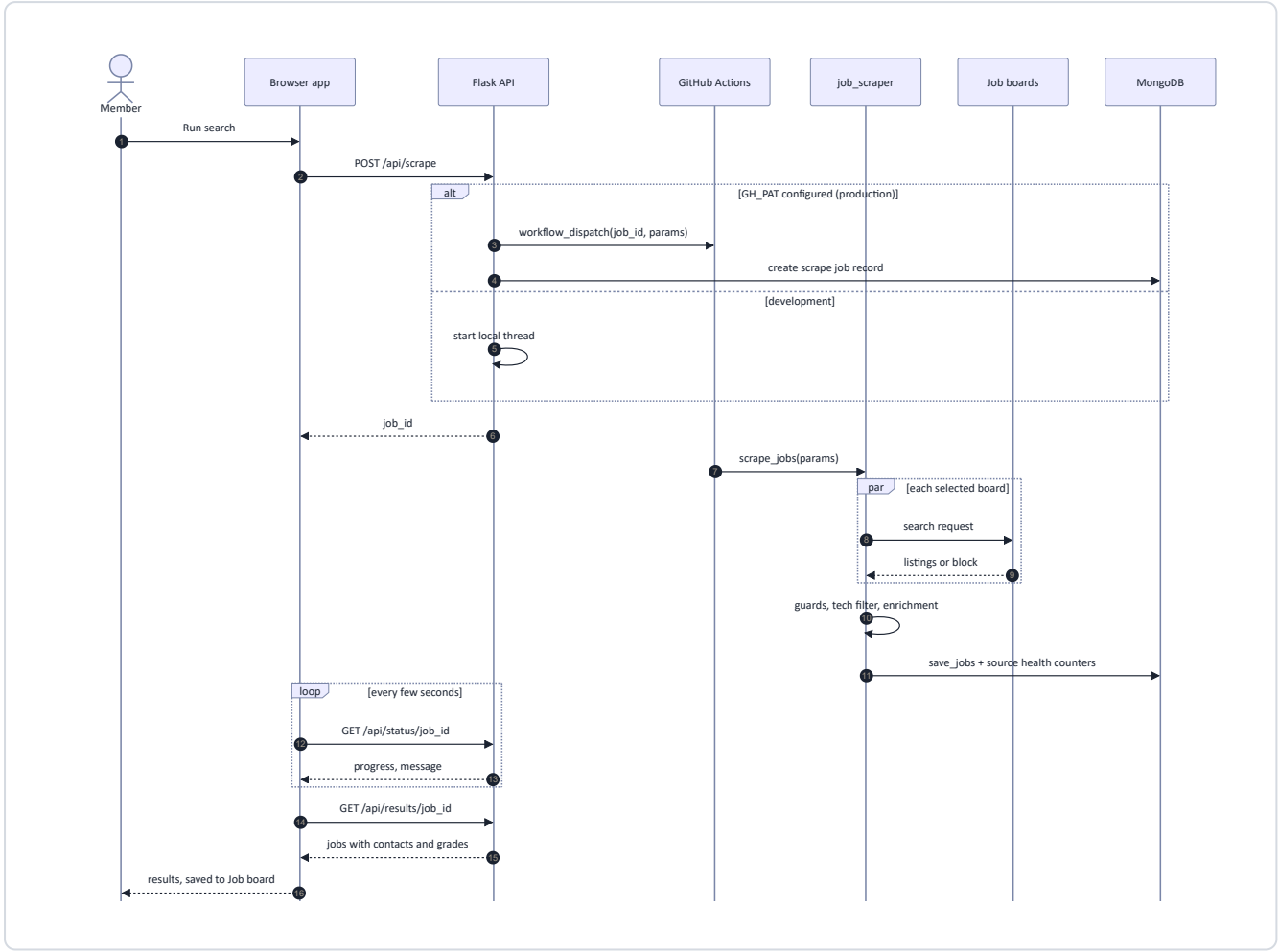
7 Data flow (level 1)

How a search becomes a ranked list of contacts. Rounded boxes are processes, cylinders are stores.



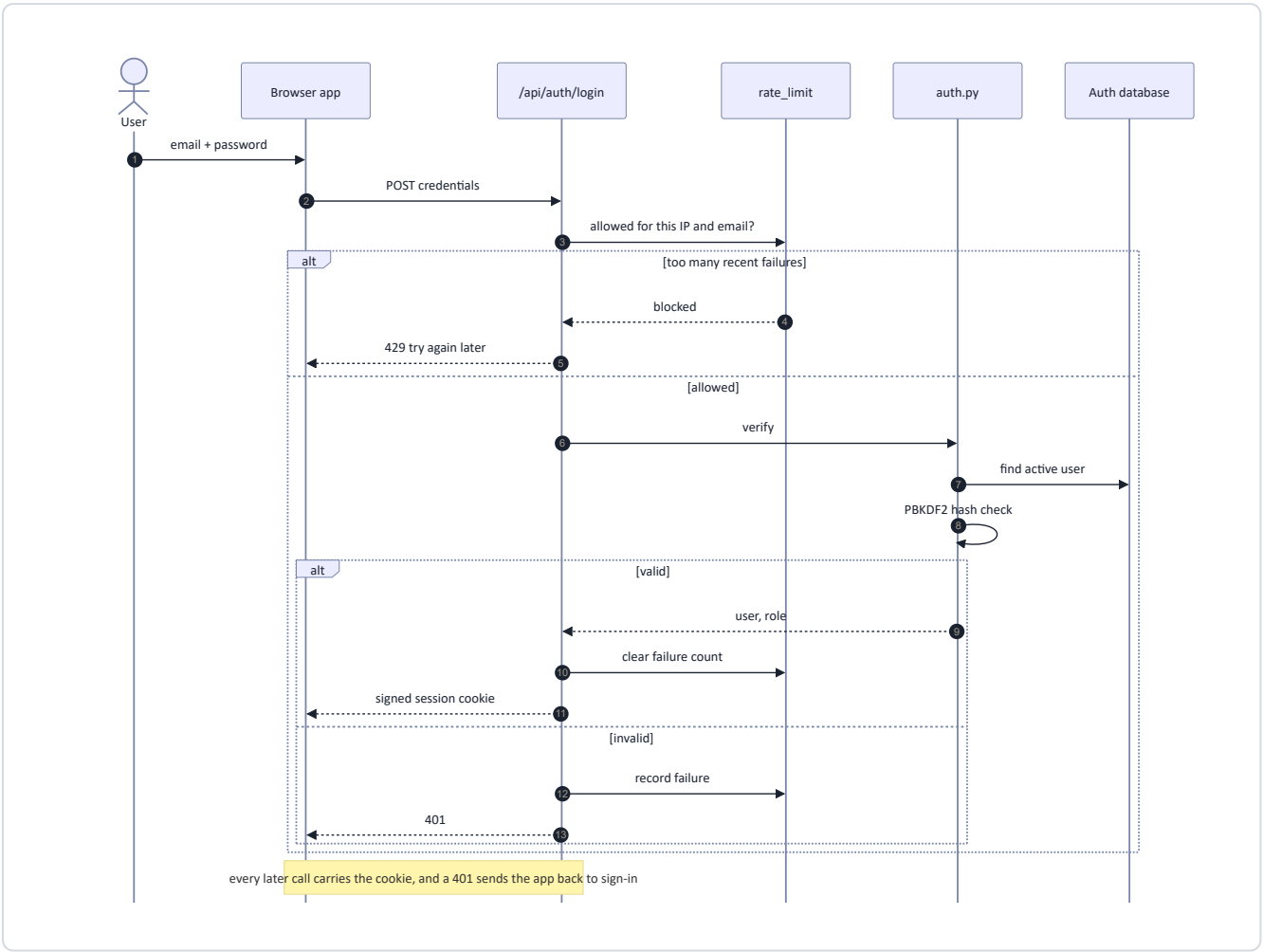
8 Sequence: job search

From pressing "Run search" to results on the Job board. The browser polls for progress while the scrape runs elsewhere.



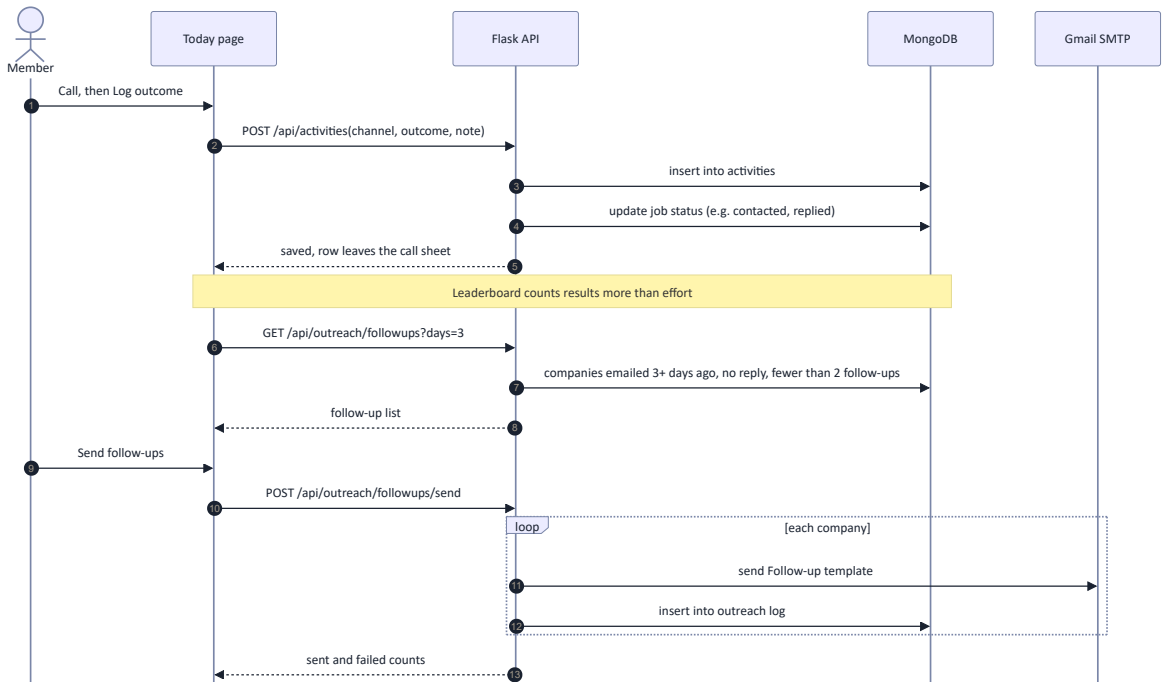
9 Sequence: sign-in

Failed attempts are counted before the password hash runs, so guessing is throttled. Accounts live in a separate auth database.



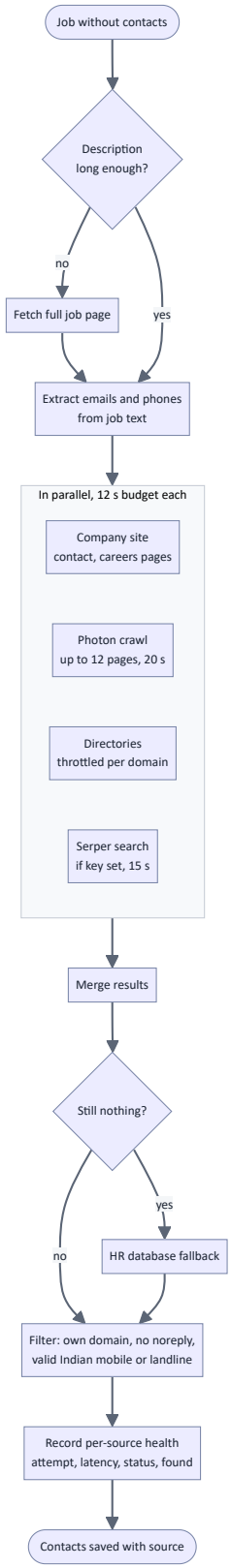
10 Sequence: logging an outcome and follow-ups

A member logs a call from the Today page; emailed companies with no reply come back as follow-ups after 3 days, at most twice.



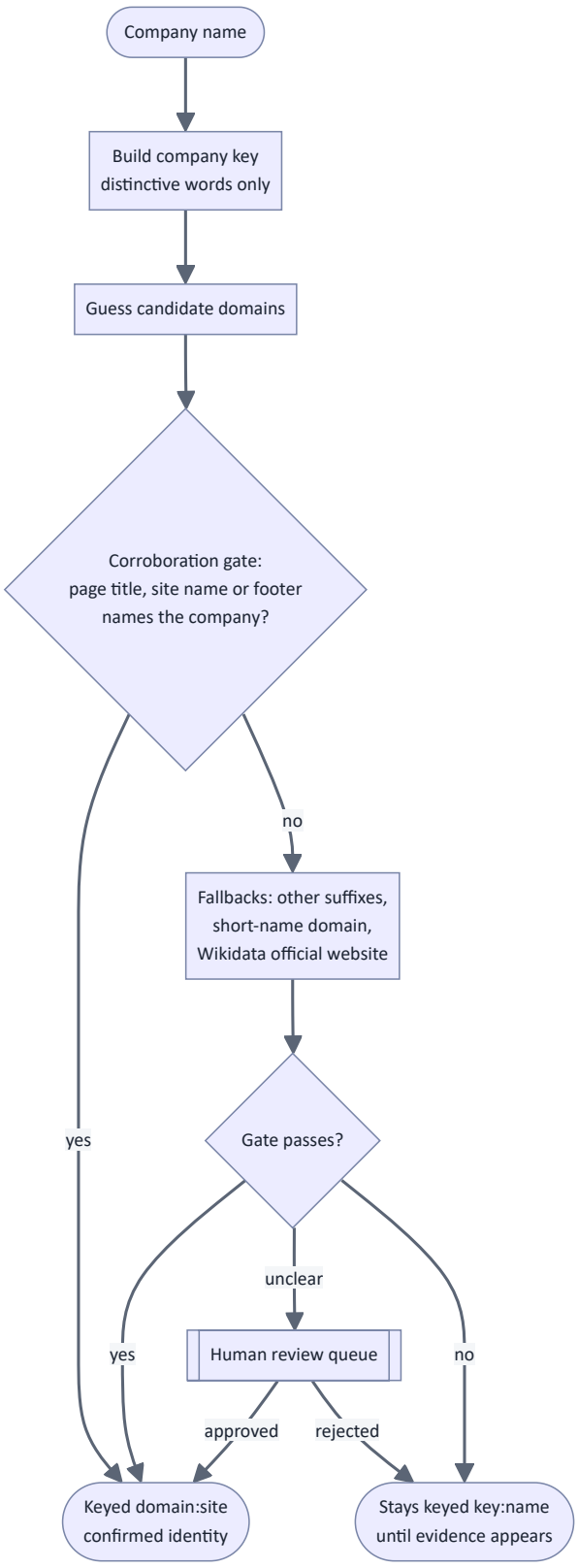
11 Activity: contact enrichment

Four jobs are enriched at once; inside each job the sources run in parallel, and each one has a fixed time to answer.



12 Activity: identity resolution

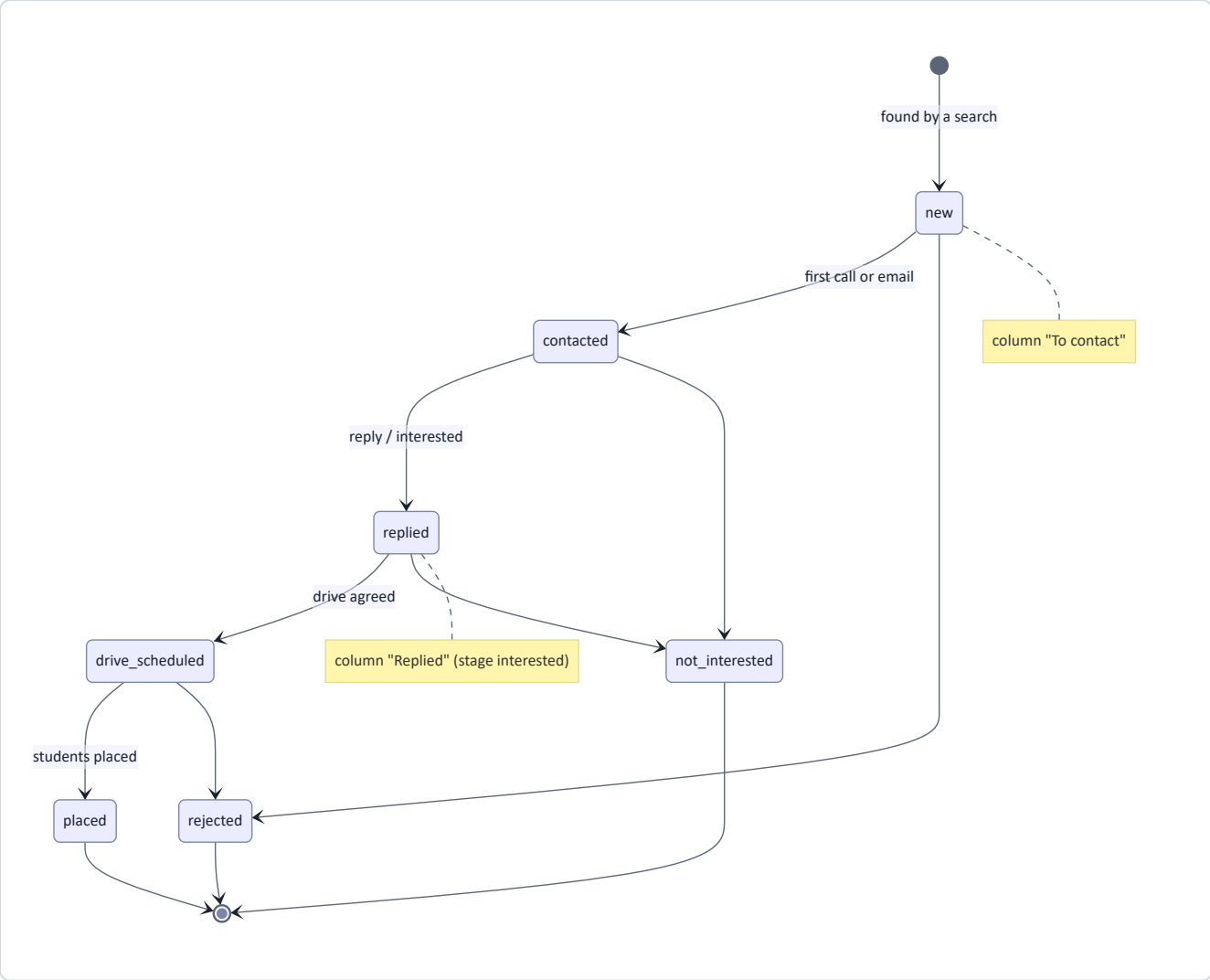
A company name alone is not an identity. A website is accepted only when the page itself names the company.



Measured result: companies with a usable website went from 21.2% to 53.3% after automated resolution and to 61.4% after human review.

13 State: company pipeline

The status a company moves through. Pipeline columns map onto these stored statuses.

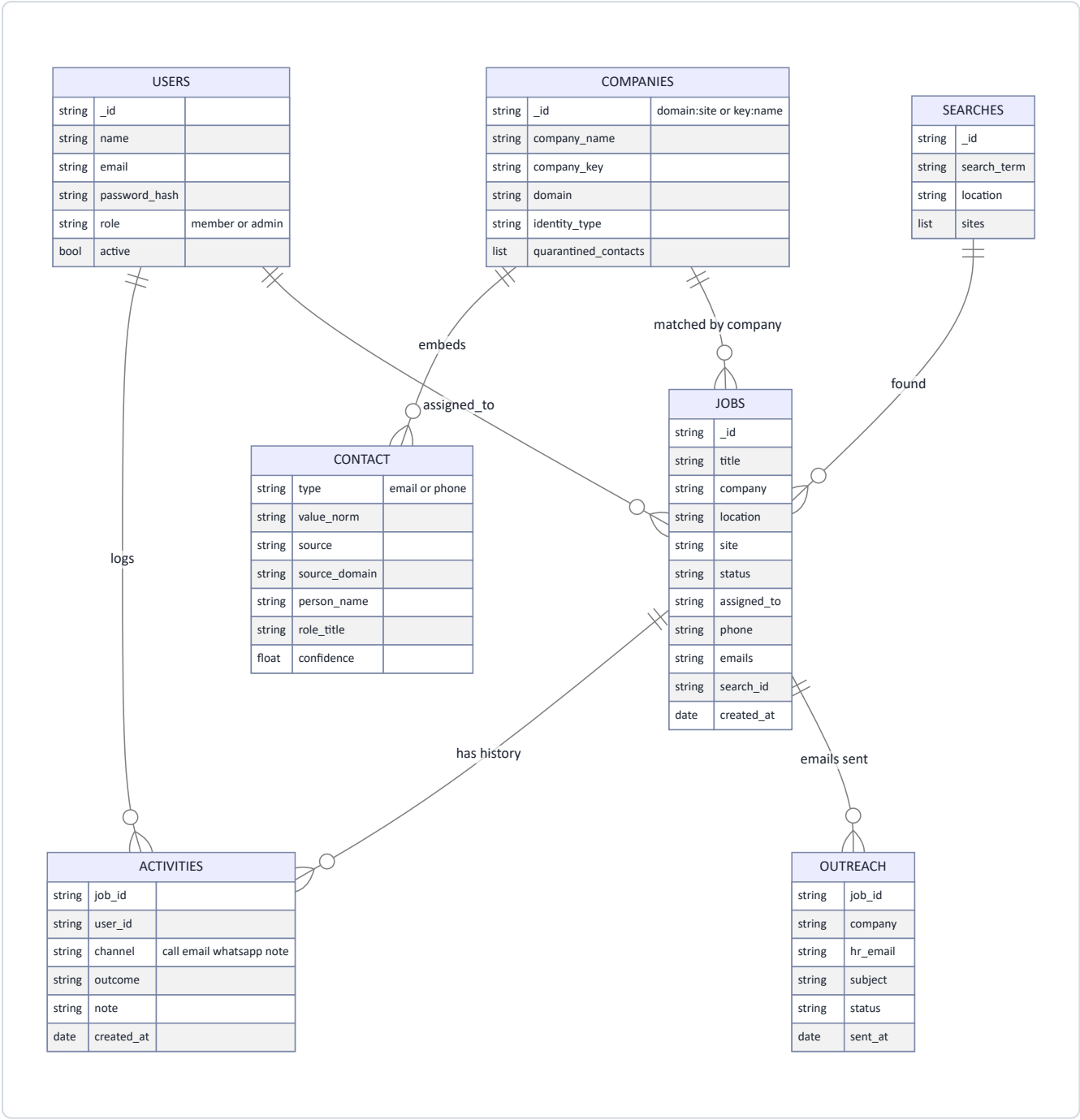


Pipeline column	Stored status
To contact	`new`
Contacted	`contacted`
Replied	`replied`
Drive scheduled	`drive_scheduled`
Placed	`placed`
Closed	`rejected`, `not_interested`

Only a company's owner or an admin can move it. Drag-and-drop and the "Move to" menu write the same status.

14 Data model (ER)

MongoDB collections and how they refer to each other. Contacts are embedded inside each company document.



Other collections: settings, custom_sites, hr_contacts, source_metrics and source_metrics_failures. Users are stored in a separate auth database.

15 Contact confidence scoring

Each contact gets a score from independent evidence, clamped to 0 to 1. The interface shows a simpler A to F grade per company.

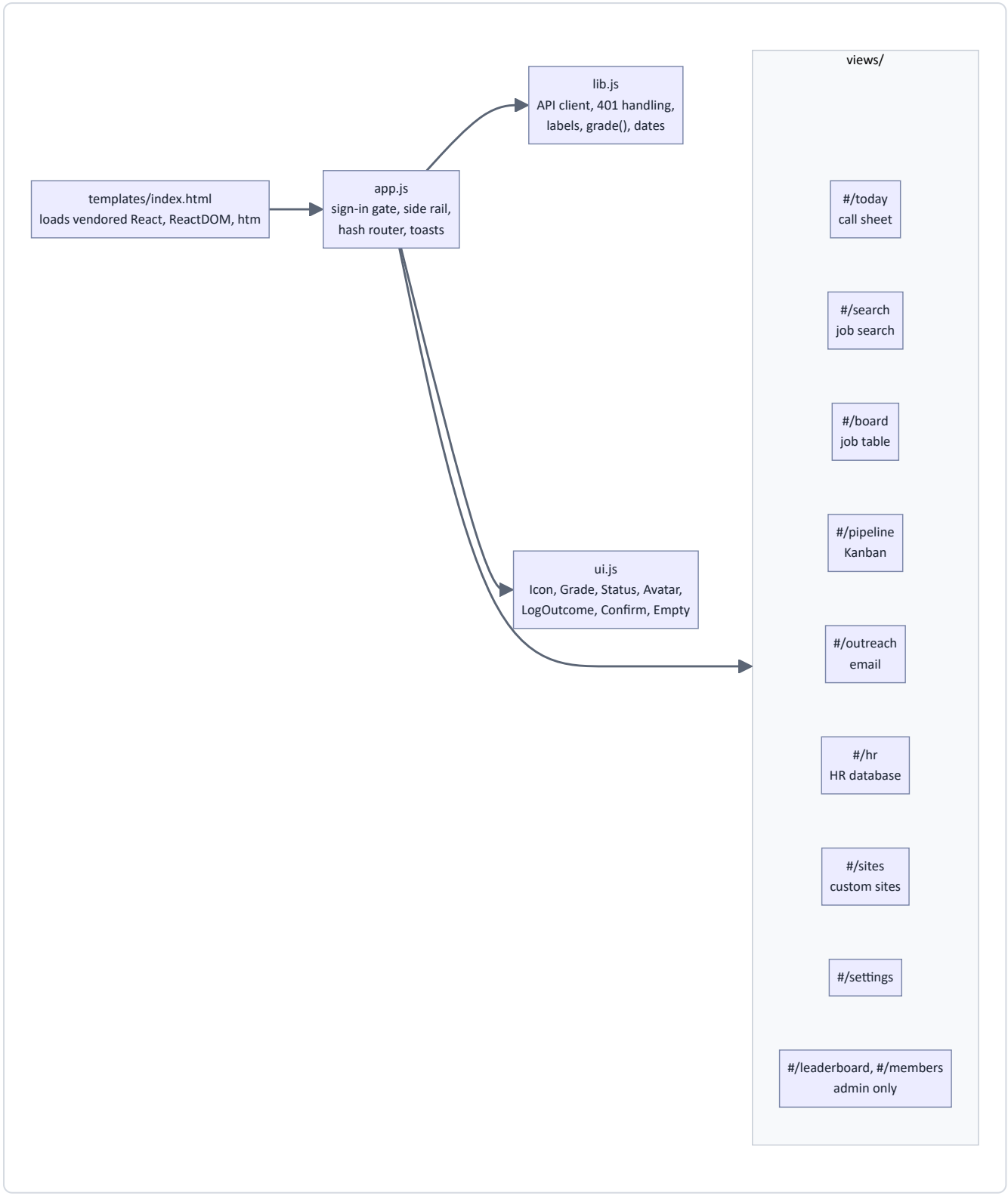
Evidence	Effect on score
Email on the company's own confirmed domain (or a phone from an ATS)	+0.30
Found through an applicant-tracking-system source	+0.25
Names a specific person	+0.20
Role title is HR, talent, recruiting, campus, people, hiring, staffing or personnel	+0.15
Mailbox verified over SMTP / only the domain's MX record checked	+0.20 / +0.05
Generic role address (info@, contact@ ...)	-0.15
Personal provider (Gmail, Yahoo ...) with no person's name	-0.20
Came from an aggregator or directory (JustDial, IndiaMART, Sulekha, TradeIndia ...)	-0.25
Same value seen on more than one company (shared switchboard)	spread penalty
Result	**Sum clamped to 0.0 to 1.0; 0.6 or above is eligible for bulk send, the rest is manual review**

Grade (company view)	Meaning	Points
A	Phone and email, with an HR-type address	80 or more
B	Phone and email	55 to 79
C	Only a phone or only an email	35 to 54
D	Company website only	15 to 34
F	No contact details yet	below 15

Grade points: phone +35, email +35, company website +15, HR-type email +15.

16 Front-end structure

The single-page app: one shell, shared helpers, and one module per view, each with its own hash link.



Below 900 px wide the side rail becomes a drawer and tables become stacked cards.

17 API map

The 40 Flask routes, grouped by purpose. Admin-only routes are marked.

Area	Routes
Auth	`POST /api/auth/login`, `POST /api/auth/logout`, `GET /api/auth/me`
Members (admin)	`GET/POST /api/users`, `DELETE /api/users/<id>`, `POST /api/users/<id>/active`
Search	`POST /api/scrape`, `GET /api/status/<job_id>`, `GET /api/results/<job_id>`, `GET /api/debug/<job_id>`
Jobs	`GET /api/jobs`, `PATCH /api/jobs/<id>`, `POST /api/jobs/enrich`, `GET /api/export`
Pipeline	`GET /api/pipeline`, `POST /api/jobs/<id>/stage`, `/assign`, `/remarks`
Activity	`POST /api/activities`, `GET /api/jobs/<id>/activities`, `GET /api/leaderboard`
Outreach	`POST /api/outreach/send`, `/bulk`, `GET /api/outreach/log`, `GET /api/outreach/followups`, `POST /api/outreach/followups/send`
HR database	`GET /api/hr-database`, `POST /api/hr-database/add`
Custom sites	`GET/POST /api/custom-sites`, `DELETE /api/custom-sites/<id>`, `/toggle`, `POST /api/custom-sites/test`
Settings (admin)	`GET/POST /api/settings`, `POST /api/settings/test-smtp`, `GET /api/source-health`
Other	`GET /` (app shell), `GET /api/dashboard` (also the health check)

18 Security layers

The checks a request or a write passes through, from the browser to the database.

